

Rajasthan **Board Class 12****Subject – Physics****Time:** $3\frac{1}{4}$ Hours**M.M: 56****Question 1:**

Write the name of physical quantity whose unit is J/C. State whether this quantity is a vector or scalar?

Solution:

It is the unit of electric potential. The work required to transfer a unit positive electric charge from an infinite distance to a given point against an electric field is known as electric potential. It is scalar quantity.

Question 2:**Define Super Conductivity****Solution:**

It is the phenomenon in which the resistance of the material to the electric current flow is zero.

Question 3:

Write the definition of figure of merit of Galvanometer.

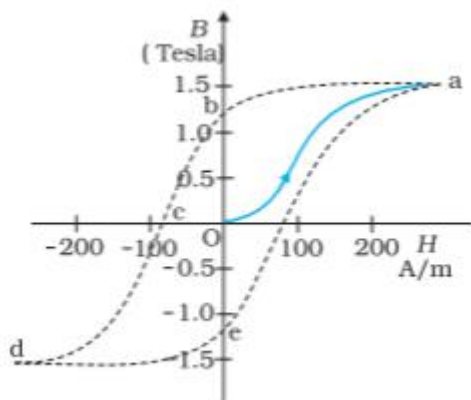
Solution:

The current required to produce a unit deflection on the galvanometer.

Question 4:

Draw hysteresis curve (B-H curve) for a ferromagnetic substance.

Solution:**B-h curve**



Question 5:

Find the time taken by alternating current to attain zero from peak value. The frequency of alternating current is 50Hz.

Solution:

WE have,

The frequency,

$$50\text{Hz}$$

$$\therefore \text{Time period} = 0.02\text{s}$$

The time taken by the alternating current to attain zero from peak is one-fourth of the time.

Now,

The time taken by alternating current to attain zero from peak value is

$$\frac{0.02}{4} \\ = 0.005$$

Question 6:

Write Malus law.

Solution:

It is termed as, if a beam of completely plane polarized light is passed through analyzer, the intensity 'I' of transmitted light varies directly as the square of the cosine of the angle θ between the transmission directions of polarizer & analyzer.

$$I = I_0 \cos^2 \theta$$

Question 7:

What does mean by work function for metals?

Solution:

The work function is defined as the minimum energy required to emit the electron from metal surface.

Question 8:

Find the de-Broglie wave length related to an electron accelerated by 10^4 volt.

Solution:

De-Broglie wavelength electron is $\frac{12.3^0}{\sqrt{V}} \text{ \AA}$

Now,

$$\begin{aligned} & \frac{12.3^0}{\sqrt{V}} \text{ \AA} \\ &= 0.123 \text{ \AA} \end{aligned}$$

Question 9:

Write the name of any one doped semiconductor used for making light emitting diode (LED).

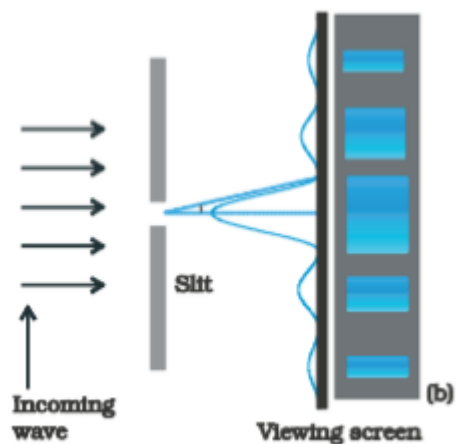
Solution:

Gallium Arsenide (GaAs), Gallium Phosphide (GaP) or Gallium Arsenide Phosphide (GaAsP) are used for making light emitting diode (LED).

Question 10:

Draw the curve for intensity distribution in diffraction by single slit.

Solution:



Question 11:

What will be the value of neutron multiplication factor for controlled chain reactions?

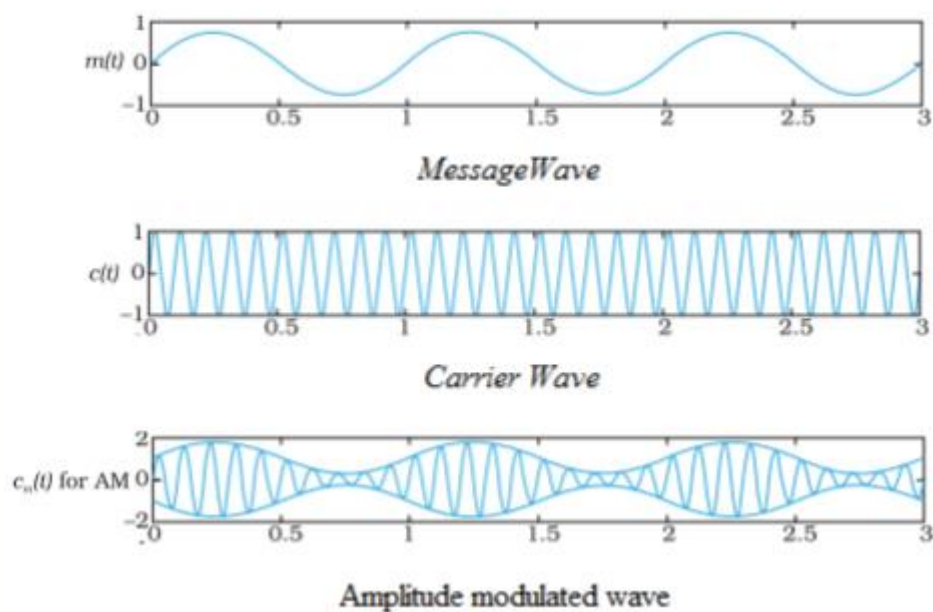
Solution:

Neutron multiplication factor in controlled chain reaction should be $k = 1$.

Question 12:

Draw a diagram of amplitude modulated wave.

Solution:



Question 13:

Write any two Maxwell's equations.

Solution:

(i) Gauss's Law $\oint E \cdot dA = \frac{Q}{\epsilon_0}$

(ii) Faraday's Law $\oint E \cdot dl = \frac{d\phi_B}{dt}$

Question 14:

Electric Potential on the surface of a charged spherical shell of radius 10cm is 50volt. Find the value of electric potential at a distance of 20cm from the centre of spherical shell.

Solution:

Electric potential on the surface of a charged spherical shell of radius r is

$$V = \frac{kq}{r}$$

$$k = \frac{1}{4\pi\epsilon_0}$$

Put $r = 0.1m, V = 50\text{volt}$ in $\frac{kq}{r}$

$$kq = 5$$

Now,

$$V = \frac{kq}{r}$$

Substitute,

$$R = 0.2m, kq = 5$$

$$V = \frac{5}{0.2}$$

$$= 25\text{volt}$$

Question 15:

The Resistance of a conductor is $X\Omega$ at 0°C temp. Find the temperature at which the resistance of conductor becomes $3X\Omega$. The temp. Coefficient of resistance for conductor is $0.4 \times 10^{-2} ^\circ\text{C}^{-1}$ which is constant.

Solution:

We have,

$$R_T = R_0 [1 + \alpha(t - t_0)]$$

R_T is the resistance at temperature T

R_0 is the resistance at temperature T_0

α is the Coefficient of resistance

Put,

$$R_T = 3\Omega, \quad R_0 = x\Omega, \quad \alpha = 0.4 \times 10^{-2} C^{-1}$$

And,

$$T_0 = 0^\circ C \text{ in } R_T = R_0 [1 + \alpha(T - T_0)]$$

$$3\Omega = x\Omega [1 + 0.4 \times 10^{-2} (T - 0)]$$

$$\Rightarrow 3 = 1 + 0.4 \times 10^{-2} T$$

$$\Rightarrow T = \frac{2}{0.4 \times 10^{-2}}$$

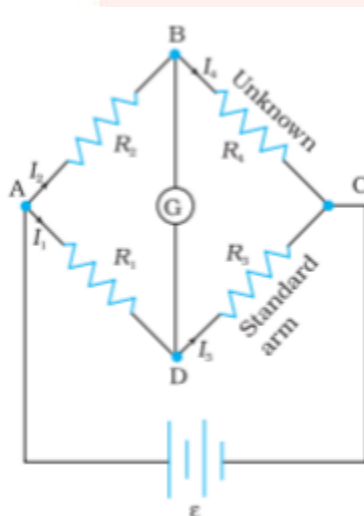
$$= 500^\circ C$$

Question 16:

Obtain necessary condition for balancing state of Wheat Stone Bridge by using Kirchoff's law.

Draw necessary circuit diagram.

Solution:



Let us consider, cell has no internal resistance. In general there will be currents flowing across all the resistors as well as a current I_g through G, the case of a balanced bridge where the resistors are such that $I_g = 0$. On applying the Kirchhoff's junction rule applied to junctions D and B, we will get $I_1 = I_3$ and $I_2 = I_4$.

Now,

$$-I_1 R_1 + 0 + I_2 R_2 = 0$$

$$\Rightarrow \frac{I_1}{I_2} = \frac{R_2}{R_1} \dots\dots(1)$$

And,

$$I_2 R_4 + 0 - I_1 R_3 = 0$$

$$\Rightarrow \frac{I_1}{I_2} = \frac{R_4}{R_3} \dots\dots(2)$$

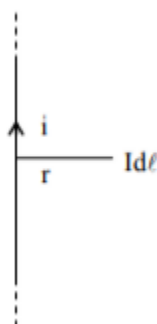
From, eq (1) and (2), we will get

$$\frac{R_2}{R_1} = \frac{R_4}{R_3}$$

Question 17:

Obtain an expression for magnetic field due to infinitely long straight current carrying conductor by using Amperes law.

Solution:



We know that,

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \sum i$$

Here,

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \sum i$$

$$\therefore B \oint dl = \mu_0 i$$

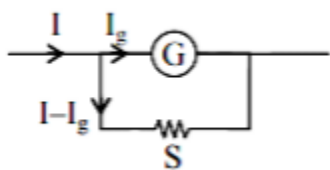
$$\Rightarrow B(2\pi r) = \mu_0 i$$

$$\Rightarrow B = \frac{\mu_0 i}{2\pi r}$$

Question 18:

The resistance of a galvanometer is 99Ω . The necessary current for full scale deflection is 1 mA . If a shunt of 1Ω is connected with galvanometer then find the value of maximum current which can be measured by this galvanometer.

Solution:



By applying Kirchhoff's loop rule we get,

$$I_g G - (I - I_g) S$$

$$\Rightarrow I = \frac{I_g G}{S} + I_g$$

$$= \frac{99 \times 10^{-3}}{1} + 10^{-3}$$

$$= 0.1\text{A}$$

Question 19:

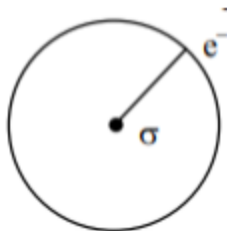
(a) Write definition of Curie temperature for ferromagnetic substances.

(b) Obtain an expression for magnetic moment of an orbital electron.

Solution:

(a) The transition temperature above which ferromagnetic substances start behaving like paramagnetic substances. It is denoted by T_c .

(b)



$$V = \frac{2\pi r}{T}$$

$$\Rightarrow T = \frac{2\pi r}{V}$$

Now,

$$I = \frac{e}{T}$$

$$I = \frac{e}{\frac{2\pi r}{V}}$$

$$\Rightarrow I = \frac{eV}{2\pi r}$$

Now,

The magnetic moment is equal the product of the current and area.

$$\therefore \mu = IA$$

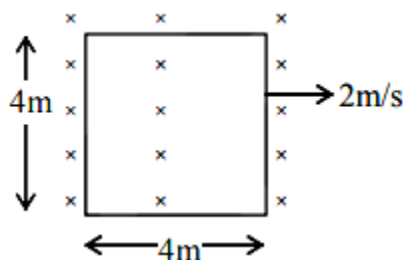
$$= \frac{eV}{2\pi r} (\pi r^2)$$

$$= \frac{eVr}{2}$$

Question 20:

The length of side of a square loop is 4m. This loop is placed in a uniform magnetic field of 2.5 t. outside the loop, the magnetic field is zero and it is coming out side from magnetic field perpendicularly with velocity 2m/s. Find the value of induced emf in loop after one second.

Solution:



We have,

The magnetic flux is $\phi = BA$

$$\begin{aligned}\Delta\phi &= \phi_2 - \phi_1 \\ &= 2.5(8-16) \\ &= -20\end{aligned}$$

Now,

$$\begin{aligned}\varepsilon &= -\frac{\Delta\phi}{dt} \\ &= -\frac{20}{1} \\ &= 20\text{volt}\end{aligned}$$

Question 21:

Find the ratio of maximum wavelength and minimum wavelength for the lines of Balmer series in hydrogen spectrum.

Solution:

$$\frac{1}{\lambda} = R\left(\frac{1}{2^2} - \frac{1}{n^2}\right)$$

And the ratio of maximum wavelength and minimum wavelength is given by

$$\begin{aligned}\frac{\frac{1}{\lambda_{\max}}}{\frac{1}{\lambda_{\min}}} &= \frac{R\left(\frac{1}{2^2} - \frac{1}{3^2}\right)}{R\left(\frac{1}{2^2} - \frac{1}{\infty^2}\right)} \\ \Rightarrow \frac{\lambda_{\max}}{\lambda_{\min}} &= \frac{\frac{5}{36}}{\frac{1}{4}} = \frac{5}{9} \\ \Rightarrow \frac{\lambda_{\max}}{\lambda_{\min}} &= \frac{9}{5}\end{aligned}$$

OR

In a radioactive sample the numbers of active nuclei remains 6.25% of its initial value in 6 hr. find the half-life of radioactive sample.

Solution:

Let us consider,

$$\frac{N}{N_0} = e^{-\lambda t}$$

$$\Rightarrow \log \frac{N}{N_0} = -\lambda t$$

$$\Rightarrow \log \frac{100}{6.25} = -\lambda t$$

$$\Rightarrow \log 2^4 = -6\lambda$$

$$\therefore \lambda = \frac{4 \log 2}{6}$$

Now,

$$T_{\frac{1}{2}} = \frac{\log 2}{\lambda}$$

$$= \frac{\log 2}{4 \log 2}$$

$$= 1.5 \text{ hours}$$

Question 22:

Derive Einstein's photoelectric equation. Explain photoelectric effect with help of this equation.

Solution:

A/C to the law of conservation of energy

$$E = \max^m \text{ K.E of } e^- + \text{Work function}$$

$$h\nu = \frac{1}{2} m V_{\max}^2 + W_0$$

$$\Rightarrow K_{\max} = h\nu - h\nu_0 \quad \left[\because W_0 = \frac{h\nu_0}{\text{Threshold frequency}} \right]$$

$$\Rightarrow K_{\max} = (v - v_0)$$

- (1) Increase of intensity means increase in the number of photons striking the metal surface at unit time.
- (2) If $v < v_0$ it is the frequency of incident radiation is less than threshold frequency. Such that the K.E of photoelectron is negative.
- (3) If $v > v_0$ the maximum kinetic energy of e^- increases linearly with frequency v .
- (4) The result of elastic collision b/w photon & e^- . The transfer of energy of photon to e^- is one lump not continuous absorption is known as photoelectric emission.

Question 23:

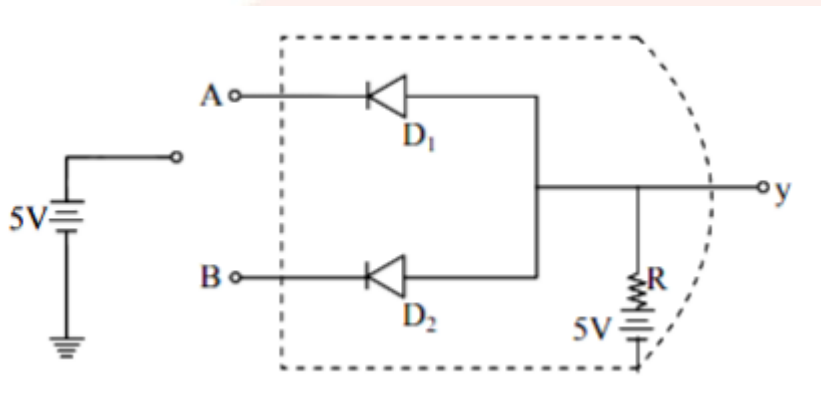
(a) What are logic gates?

Sol:

It is the fundamental building blocks of digital integrated circuit which performs basic logic functions.

(b) Draw a circuit diagram for dual-input AND Gate by using two diodes.

Solution:

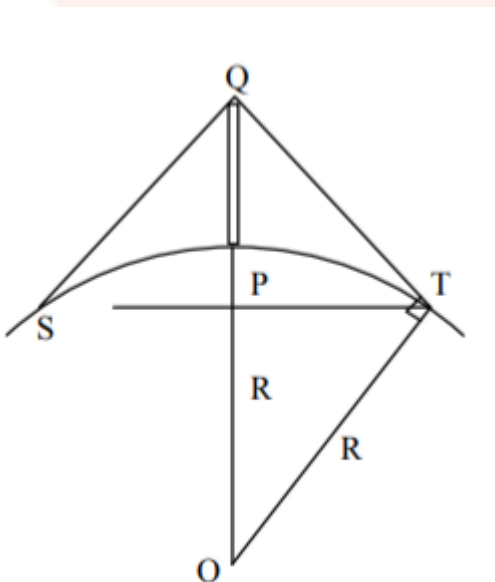


Question 24:

Establish a relation between maximum distance of broadcast and height of antenna in space wave

Propagation.

Solution:



Let us consider R is the radius of Earth and height of antenna $PQ = h_T$

Now,

$$PS = PT = QT = d$$

Therefore, in right triangle, OTQ

$$OQ^2 = OT^2 + QT^2$$

$$(R + h_T)^2 = R^2 + d^2$$

$$\Rightarrow d^2 = 2Rh_T \left(1 + \frac{h_T}{2R} \right)$$

$$\Rightarrow d^2 = 2Rh_T \left[\frac{h_T}{2R} \ll 1 \right]$$

$$\Rightarrow d = \sqrt{2Rh_T}$$

Thus, the maximum distance between transmitting & receiving antennas is

$$d_M = d_T + d_R$$

$$\Rightarrow d = \sqrt{2Rh_T} + \sqrt{2Rh_R}$$

Question 25:

(a) What does mean by magnifying power of a Microscope?

Sol:

The ratio of the angle subtended at the eye by the image to the angle subtended at the eye by the object, when both are at the least distance of distinct vision from the eye is known as the magnifying power.

(b) An object is placed at 20 cm from a convex lens. If 3 times magnified real image is formed by the lens then find the focal length of lens.

Solution:

We have,

$$d_M = d_T + d_R$$

$$\Rightarrow d = \sqrt{2Rh_T} + \sqrt{2Rh_R}$$

Question 26:

What does mean by mass defect? Establish relation between mass defect and nuclear binding energy. And hence write the expression for binding energy per nucleon.

Solution:

The difference in mass of a nucleus and its constituents is called the mass defect.

$$\Delta M = [Zm_p + (A - Z)m_n] - M$$

The mass of the oxygen nucleus is less than the sum of the masses of its constituents (8 protons and 8 neutrons, in the unbound state), the equivalent energy of the oxygen nucleus is less than that of the sum of the equivalent energies of its constituents. Extra energy ΔMc^2 is applied to break the oxygen nucleus into 8 protons and 8 neutrons. This energy is known as the binding energy.

$$E_n = \Delta Mc^2$$

$$= [Zm_p + (A - Z)m_n - M]c^2$$

Since, the binding energy per nucleon E_{bn} is the ratio of the binding E_b of nucleus to the number of the nucleons.

$$E_{bn} = \frac{E_b}{A}$$

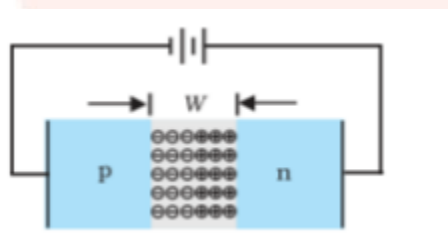
Question 27:

Draw circuit diagram for a P-N junction to obtain reverse bias characteristic curves. Explain the phenomenon of reverse breakdown for a P-N junction in reverse bias state by following processes –

(i) Avalanche breakdown

(ii) Zener breakdown

Solution:



If the reverse voltage will increase then the depletion layer breaks at certain reverse voltage, it is known as break down voltage.

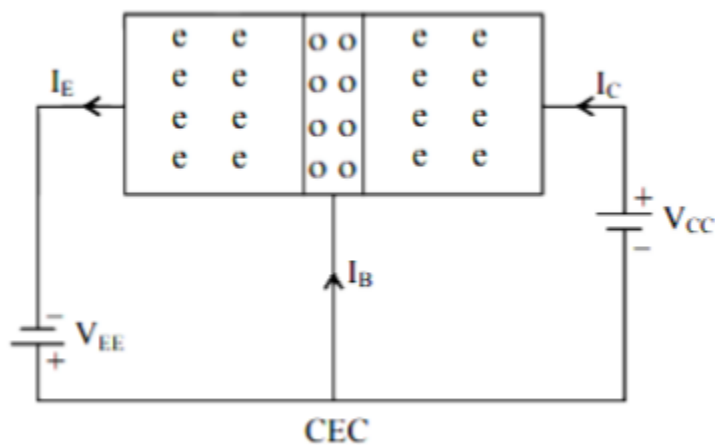
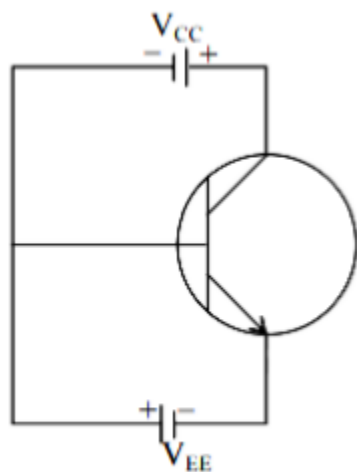
Avalanche breakdown: - It is defined as the process of applying high voltage and increasing the free electrons or electric current in semiconductors and insulating materials.

Zener breakdown: - It is defined as the process in which the electrons move across the barrier from the valence band of p-type material to the conduction band of n-type material.

OR

Draw circuit diagram for a NPN-transistor to obtain characteristic curves in common base configuration. Establish a relation between current amplification factors for a transistor in common base and common emitter configuration.

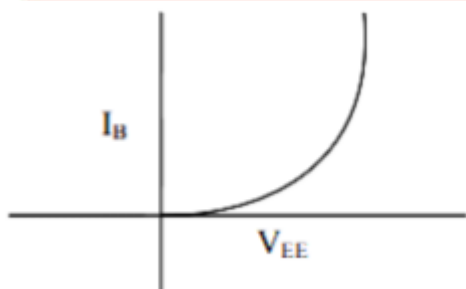
Solution:



So,

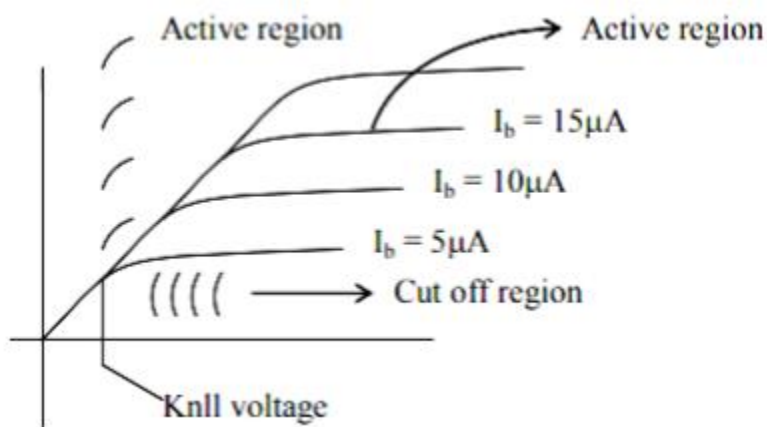
Input Characteristics $V_{cc} = \text{constant}$

V_{EE} is increases



And the Characteristics V_{cc} is increases

$V_{EE} = \text{constant}$



Now, the current amplification for the common emitter and common base configuration

$$\alpha = \frac{I_C}{I_E} \text{ and } \beta = \frac{I_C}{I_B}$$

Then,

$$\alpha = \frac{I_C}{I_E}$$

$$\Rightarrow \alpha = \frac{\frac{I_C}{I_B}}{\frac{I_E}{I_B}}$$

$$\Rightarrow \alpha = \frac{\frac{I_C}{I_B}}{\frac{I_B + I_C}{I_B}}$$

$$\Rightarrow \alpha = \frac{\frac{I_C}{I_B}}{1 + \frac{I_C}{I_B}}$$

$$\Rightarrow \alpha = \frac{\beta}{1 + \beta} \left[\because \beta = \frac{I_C}{I_B} \right]$$

Question 28:

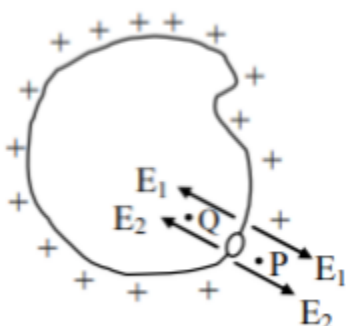
Write definition of electric field intensity. Obtain an expression for electric force and electric pressure on the surface of a charged conductor. Draw necessary diagram.

Solution:

The strength of an electric field at any point. It is equal to the electric force per unit charge experienced by a test charge placed at that point is known as the electric field intensity. Its SI unit is Newton/Coulomb.

Since, every element of a charged conductor experience a normal outward mechanical force. It is the result of repulsive force from similar charges present on the rest of the surface of the conductor. If a small element ds on the surface of a charge conductor. if σ is the surface charge density and the charge carried by the element ds be dq .

$$\therefore dq = \sigma \cdot ds$$



Now, consider a point P just outside the surface near the element ds .

$$E = \frac{\sigma}{\epsilon_0}$$

The direction of the intensity is normally outwards. It is made up of two components $\vec{E}_1$ due to the small charge dq . $\vec{E}_1$ Due to the remaining charge present on the rest of the surface of the conductor.

Therefore,

$$\text{at P, } \vec{E}_1 + \vec{E}_2 = \frac{\sigma}{\epsilon_0}$$

Also, consider a point Q inside the conductor the electric field E_1 and E_2 . Since, the intensity inside the charged conductor is zero.

$$\vec{E}_1 - \vec{E}_2 = 0$$

$$\Rightarrow \vec{E}_1 = \vec{E}_2$$

Now,

$$\vec{E}_1 + \vec{E}_2 = \frac{\sigma}{\epsilon_0}$$

$$\Rightarrow 2\vec{E}_2 = \frac{\sigma}{\epsilon_0}$$

$$\Rightarrow \vec{E}_2 = \frac{\sigma}{2\epsilon_0}$$

Here, E_2 is the electric field due to the charge on the rest of conductor.

$$\Rightarrow \vec{F} = \vec{E}_2 \cdot dq$$

$$= \frac{\sigma}{2\epsilon_0} \cdot \sigma ds \quad [\because dq = \sigma ds]$$

$$= \frac{\sigma^2}{2\epsilon_0} ds$$

Such that the electric pressure on the surface is

$$P = \frac{\frac{\sigma^2}{2\epsilon_0} ds}{ds}$$

$$\Rightarrow P = \frac{\sigma^2}{2\epsilon_0} \text{ N / m}^2$$

OR

What does mean by energy stored in capacitor?

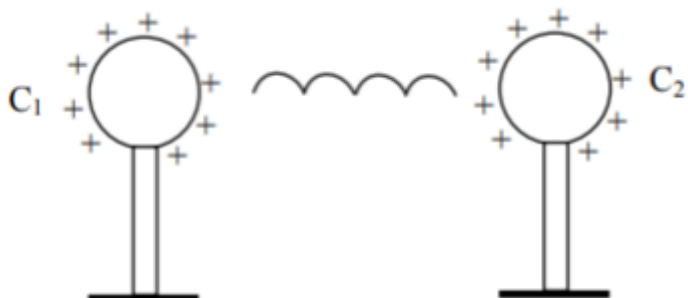
Prove that – "The ratio of change in potentials of conductors after connecting two charged conductors is inversely proportional to their capacitances". Draw a curve between electric field and distance for and uniformly for a uniformly charged non-conducting sphere.

Solution:

The process of charging up a capacitor involves the transferring of electric charges from its one plate to another. The work done in charging the capacitor is stored as its electrical potential energy is known as the energy stored in a capacitor.

Let us suppose that two insulated conductors A and B of capacitance C_1 and C_2 and carrying charges Q_1 and Q_2 .

$$Q_1 = C_1 V_1 \text{ and } Q_2 = C_2 V_2$$



If connected with wire, charge flows from higher to lower potential.

Total charge on system $Q_1 + Q_2$

Total capacitance $C_1 + C_2$

$$V = \frac{Q}{C}$$

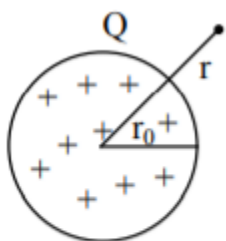
$$\Rightarrow V = \frac{Q_1 + Q_2}{C_1 + C_2}$$

$$\Rightarrow V = \frac{C_1 V_1 + C_2 V_2}{C_1 + C_2}$$

$$V \propto \frac{1}{C_1 + C_2}$$

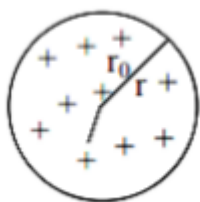
Now, the electric field distance curve by uniformly charged non-conducting sphere.

$$E = \frac{KQ}{r^2}$$

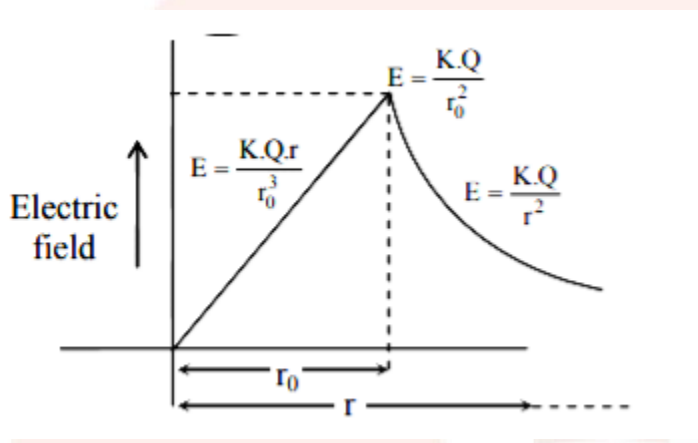


E inside ($r < r_0$)

$$E = \frac{KQ}{r_0^2}$$



Therefore, the curve between electric field and the distance for uniformly charged conducting sphere is



Question 29:

Write one merit and one de-merit of alternating current in comparison of direct current.

Obtain expression for following in a pure inductive alternating current circuit –

- (i) Instantaneous value of current**
- (ii) Reactance of circuit**
- (iii) Peak value of current**

Draw curve for power in pure inductive circuit.

Solution:

Comparison of direct current is that the generation of AC is more economical than DC. It is said on merits. While, on demerits alternating current in comparison of direct current is that the it is dangerous to work with AC.



Now, by using Kirchhoff's loop rule,

$$\sum \varepsilon(t) = 0$$

Since, there is no any resistor in the circuit,

$$\therefore v - L \frac{di}{dt} = 0$$

Where, the second term is the self-induced Faraday emf in the inductor; and L is the selfinductance the inductor.

$$\frac{di}{dt} = \frac{v}{L}$$

$$\Rightarrow \frac{di}{dt} = \frac{v_0 \sin \omega t}{L}$$

$$\Rightarrow \int \frac{di}{dt} dt = \frac{v_0}{L} \int \sin \omega t dt$$

$$\Rightarrow i = \frac{v_0}{\omega L} \cos \omega t + C$$

$$\Rightarrow i = \frac{v_0}{X_L} \sin \left(\omega t - \frac{\pi}{2} \right) + C$$

$$\Rightarrow i = i_0 \sin \left(\omega t - \frac{\pi}{2} \right) + C$$

The integration constant has the dimension of current and is time independent. Since the source has an emf which oscillates symmetrically about zero. Such that the constant of integration is zero.

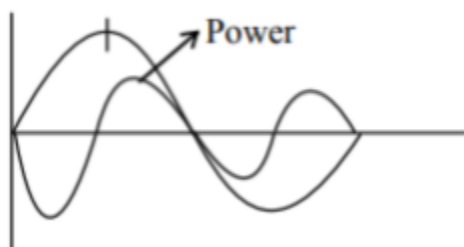
$$\therefore i = i_0 \sin \left(\omega t - \frac{\pi}{2} \right)$$

$$(i) \quad \therefore i_0 = \frac{v_0}{X_L} = \frac{v_0}{\omega L}$$

$$(ii) \quad X_L = \omega L$$

$$(iii) \quad i = i_0 \sin\left(\omega t - \frac{\pi}{2}\right)$$

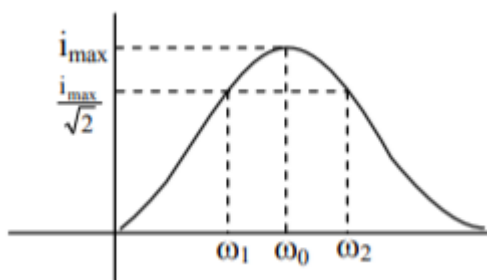
The curve is



OR

What does mean by half power point frequencies? Obtain an expression for bandwidth in a LCR series circuit. Show half power point frequencies in curve between alternating current and frequency.

Solution:



We have seen in the graph

$$\omega_1 = \omega_0 - \Delta\omega$$

$$\omega_2 = \omega_0 + \Delta\omega$$

Then,

$$\omega_2 - \omega_1 = 2\Delta\omega$$

The ω_1 and ω_2 has frequency at which maximum current becomes

$$\frac{1}{\sqrt{2}} \text{ Times.}$$

Now,

$$\begin{aligned}
 V_0 &= I_0 Z \\
 \Rightarrow \frac{V_0}{Z} &= \frac{I_0}{\sqrt{2}} \\
 \Rightarrow \frac{I_0 R}{Z} &= \frac{I_0}{\sqrt{2}} \\
 \Rightarrow Z &= \sqrt{2} R \\
 \Rightarrow \sqrt{R^2 + (X_C - X_L)^2} &= \sqrt{2} R \\
 \Rightarrow R^2 + \left(\frac{1}{\omega C} - \omega L \right)^2 &= 2R^2 \\
 \Rightarrow R &= \frac{1}{\omega C} - \omega L \\
 \Rightarrow R &= \frac{1}{\omega_1 C} - \omega_1 L \\
 \Rightarrow R &= \frac{1}{(\omega - \Delta\omega)C} - (\omega - \Delta\omega)L \\
 \Rightarrow R &= \frac{1}{\omega_0 \left(1 - \frac{\Delta\omega}{\omega_0} \right) C} - \omega_0 \left(1 - \frac{\Delta\omega}{\omega_0} \right) L \\
 \Rightarrow R &= \omega_0 L \left(1 + \frac{\Delta\omega}{\omega_0} - 1 + \frac{\Delta\omega}{\omega_0} \right) \\
 \Rightarrow R &= \omega_0 L \left(\frac{2\Delta\omega}{\omega_0} \right) \\
 \Rightarrow \Delta\omega &= \frac{R}{2L}
 \end{aligned}$$

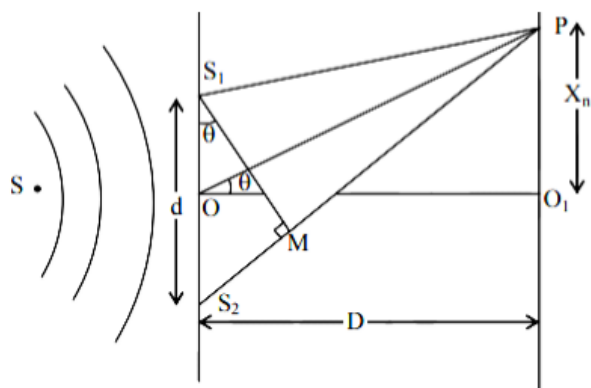
Question 30:

Write any two necessary conditions for interference of light. Obtain an expression for fringe width in Young's double slit experiment. Draw curve for intensity distribution in Young's double slit experiment.

Solution:

The two necessary condition are as follows:

- (a) Light should be propagate in same direction
- (b) Light should be coherent lights.



In $\Delta S_1 S_2 M$

$$\sin \theta = \frac{S_2 M}{d}$$

$$\Rightarrow \theta = \frac{S_2 M}{d} \dots (1)$$

In $\Delta O O_1 P$

$$\sin \theta = \frac{X_n}{D}$$

$$\Rightarrow \theta = \frac{X_n}{D} \dots (2)$$

From eq (1) and (2) we get,

$$\frac{S_2 M}{d} = \frac{X_n}{D}$$

$$\Rightarrow S_2 M = \frac{d X_n}{D}$$

$$\Rightarrow \frac{X_n}{D} = n \lambda$$

$$\Rightarrow X_n = \frac{n \lambda D}{d}$$

For dark fringe width,

$$\beta_{\text{dark}} = X_{n+1} - X_n$$

$$\Rightarrow \beta_{\text{dark}} = \frac{(n+1) \lambda D}{d} - \frac{n \lambda D}{d}$$

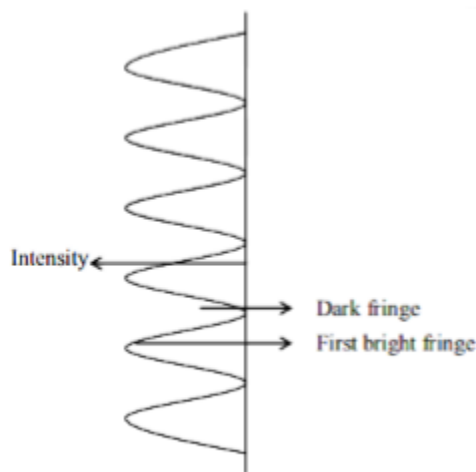
$$\Rightarrow \beta_{\text{dark}} = \frac{\lambda D}{d}$$

For bright fringes width,

$$\beta_{\text{bright}} = X_{n+1} - X_n$$

$$\Rightarrow \beta_{\text{bright}} = \frac{[2(n+1)+1]\lambda D}{d} - \frac{(2n+1)\lambda D}{d} \quad \left[\because \frac{X_n}{D} = \frac{(2n+1)\lambda D}{d} \right]$$

$$\Rightarrow \beta_{\text{bright}} = \frac{\lambda D}{d}$$



OR

Write definitions of plane of vibration and plane of polarization. Explain the working process to obtain plane polarized light by Nicole Prism. Draw necessary diagram.

Solution:

The plane containing the optic axis in which the vibrations occur is known as plane of vibration. Whereas, the plane which is at right angles to the plane of vibration and which contains the direction of propagation of the polarized light is known as the plane of polarization

Polarization by Nicole prism:

Principle:

If a thin film of canada balsam is placed between two calcite pieces, the Ordinary rays of the unpolarised incident light get eliminated through the phenomenon of total internal reflection while E-rays are transmitted unaffected and emerge as a beam of plane polarized light.

Working:

The principal section ACGE of a Nicole prism. The diagonal AG represents the canada balsam layer. When ray of unpolarised light passes from a portion of the calcite crystal into the layer of canada balsam, it passes from a denser to a rarer medium. When the angle of incidence is greater than the critical angle. Since, the ray is totally reflected and absorbed by _ 69° a blackened

surface. E-ray is not affected because it is travelling from rarer medium to denser medium. It gets transmitted through the Nicole prism.

